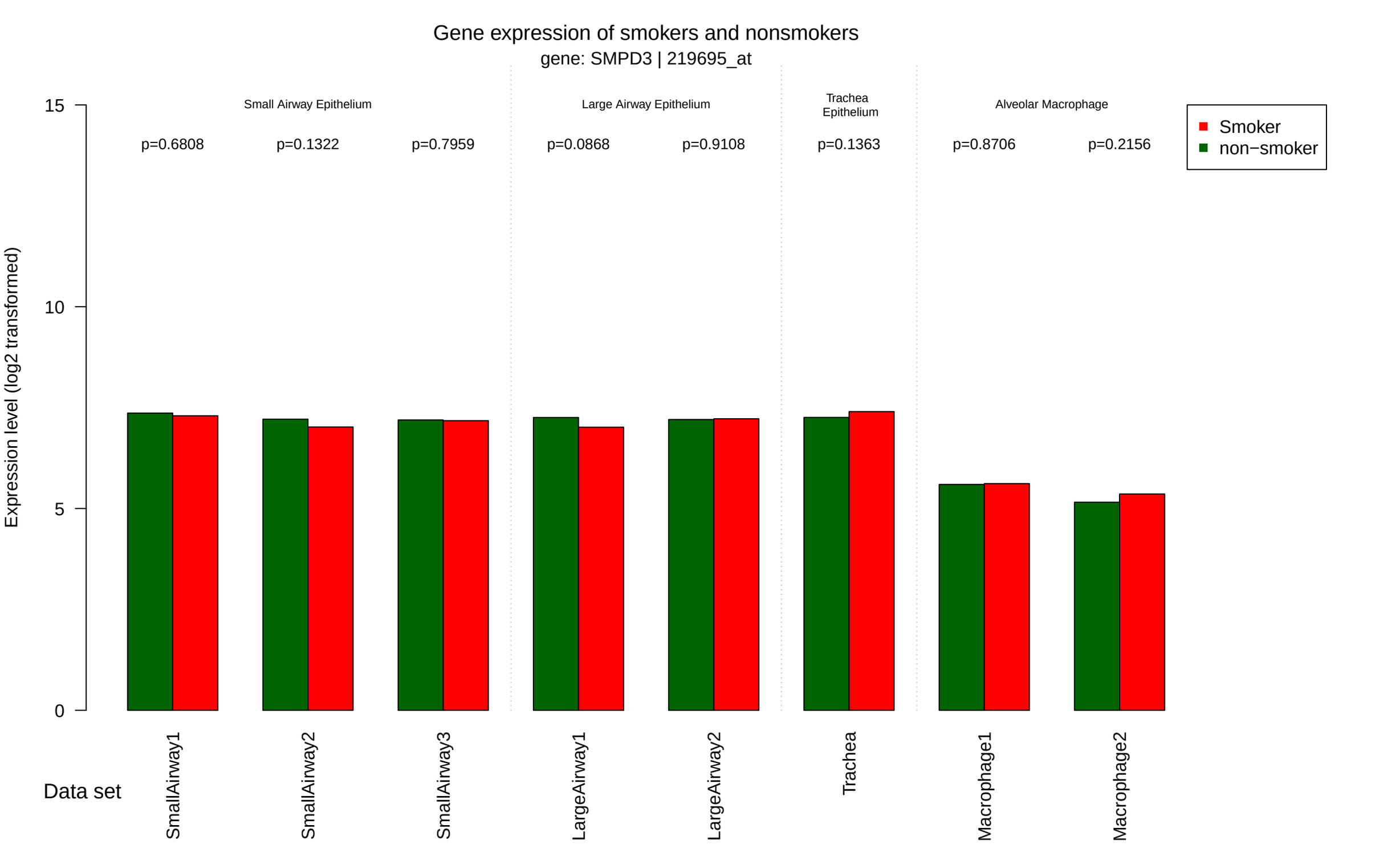
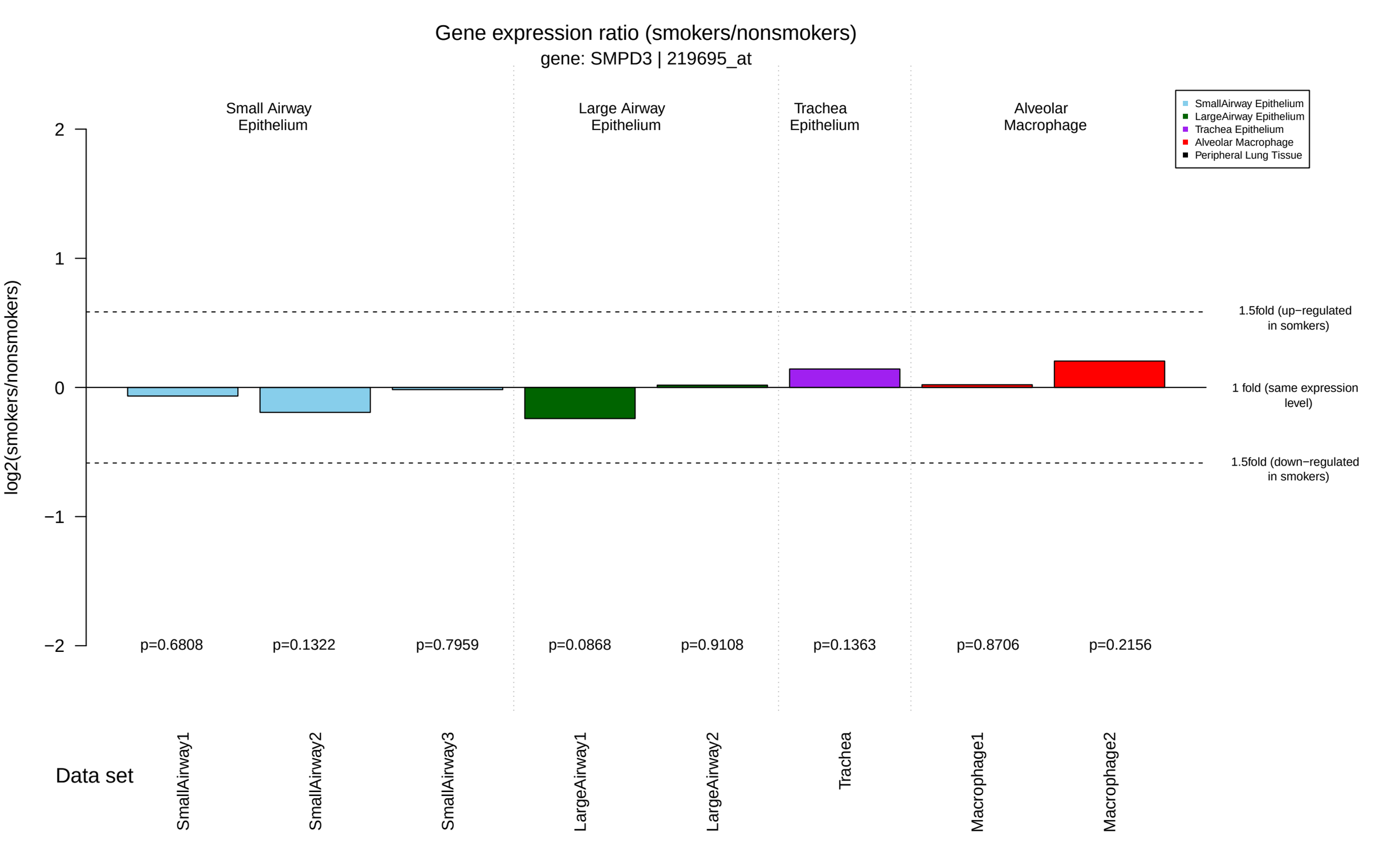
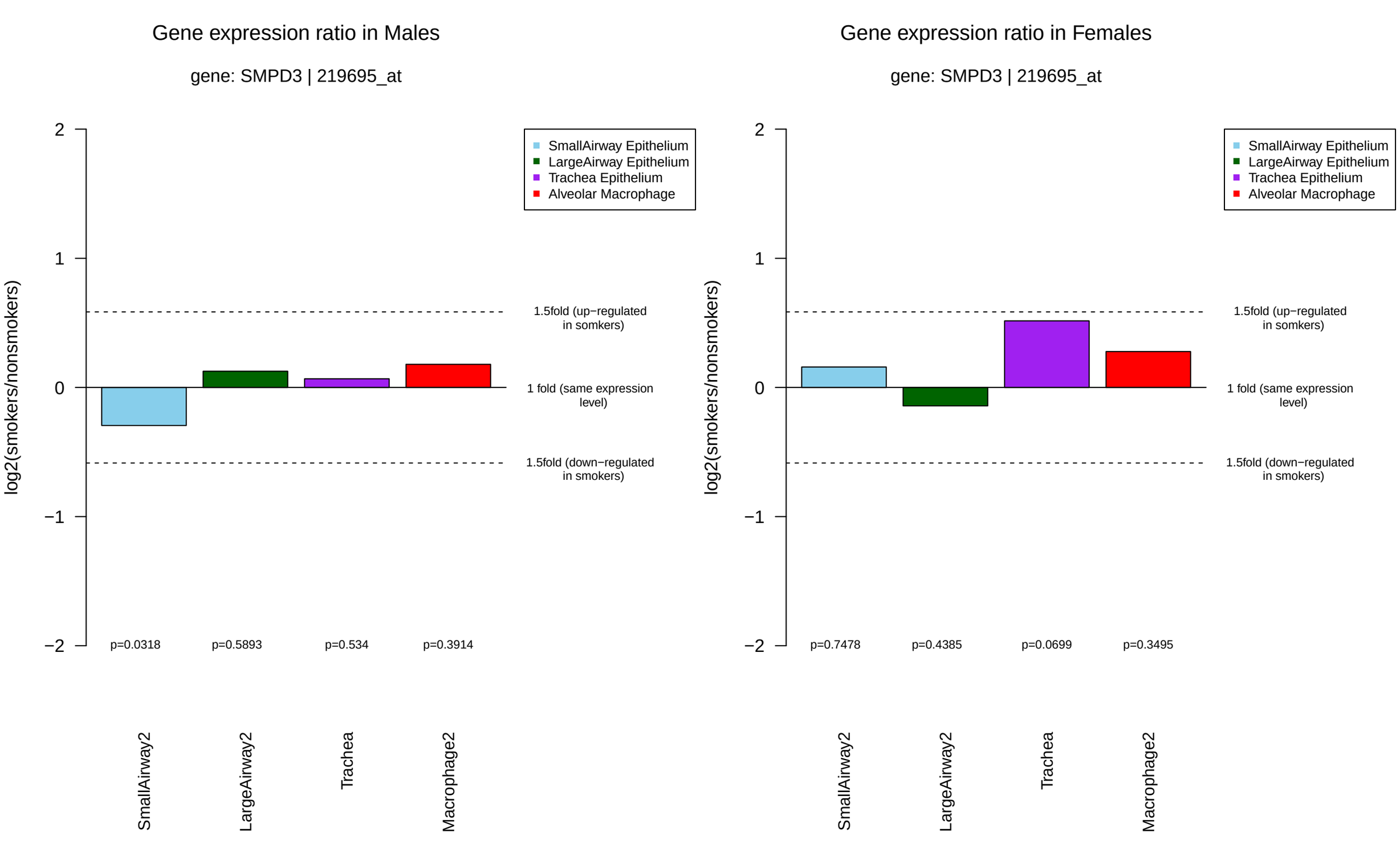
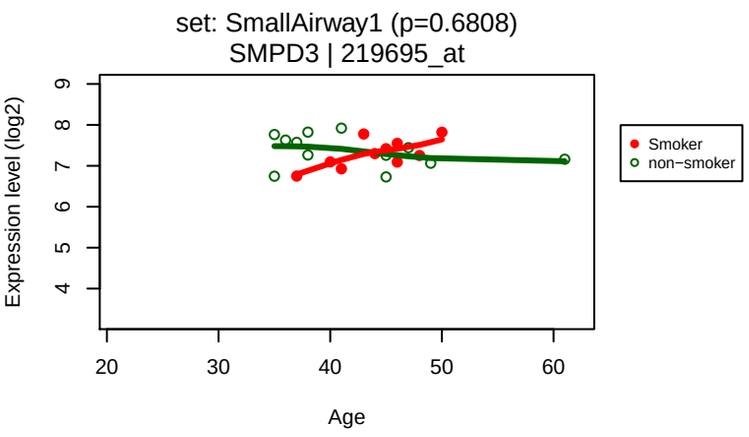
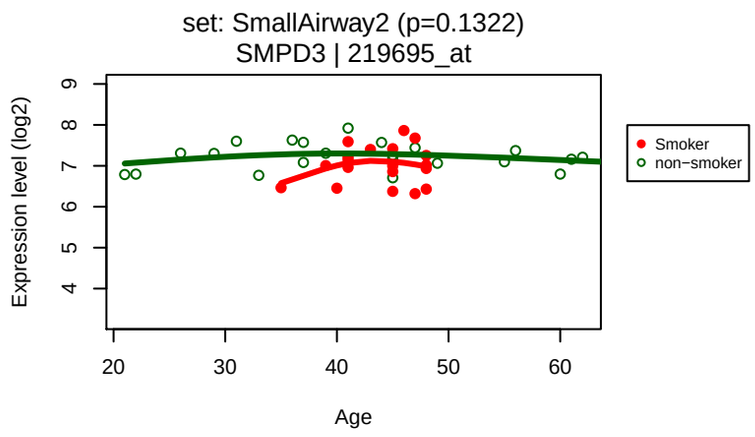
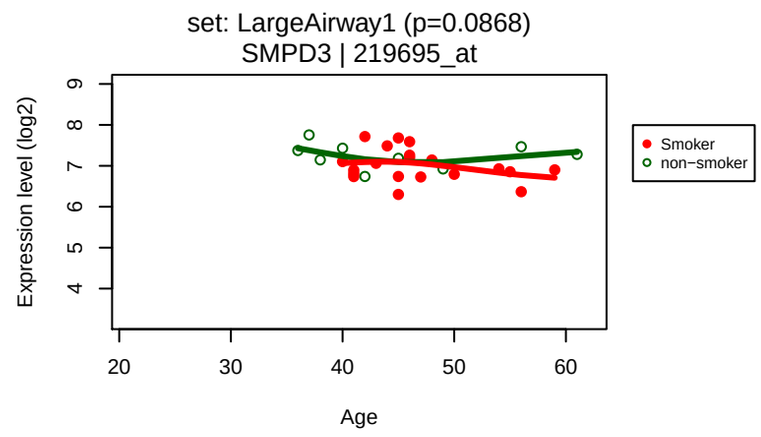

Gene expression of smokers and nonsmokers
gene: SMPD3 | 219695_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6808
p=0.1322
p=0.7959
p=0.0868
p=0.9108
p=0.1363
p=0.8706
p=0.2156
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

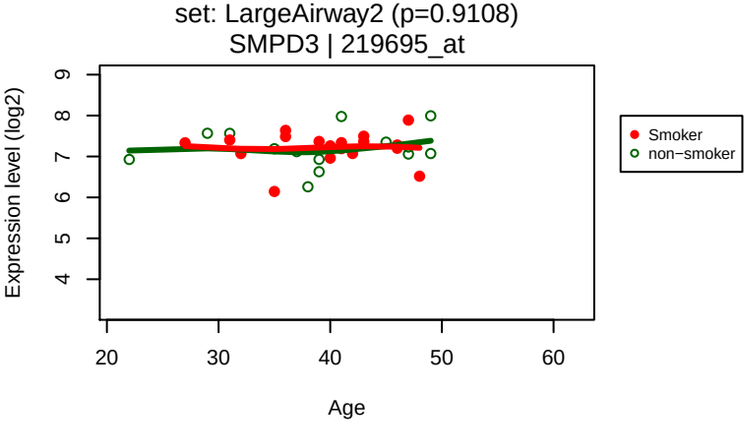

Gene expression ratio (smokers/nonsmokers)
gene: SMPD3 | 219695_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6808
p=0.1322
p=0.7959
p=0.0868
p=0.9108
p=0.1363
p=0.8706
p=0.2156
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

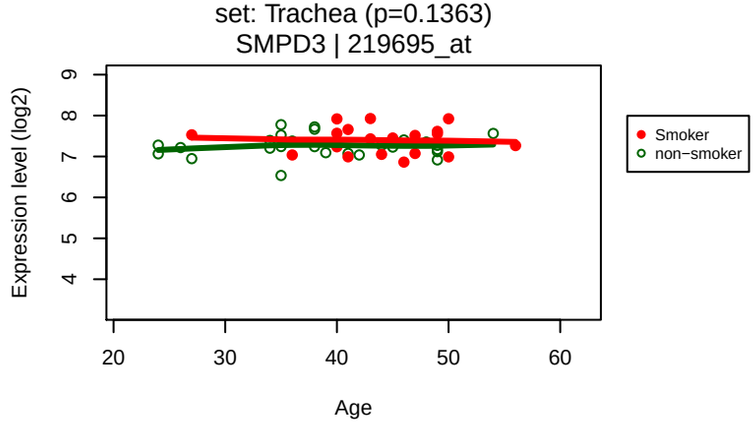

Gene expression ratio in Males
gene: SMPD3 | 219695_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0318
p=0.5893
p=0.534
p=0.3914
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMPD3 | 219695_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7478
p=0.4385
p=0.0699
p=0.3495
SmallAirway2
LargeAirway2
Trachea
Macrophage2

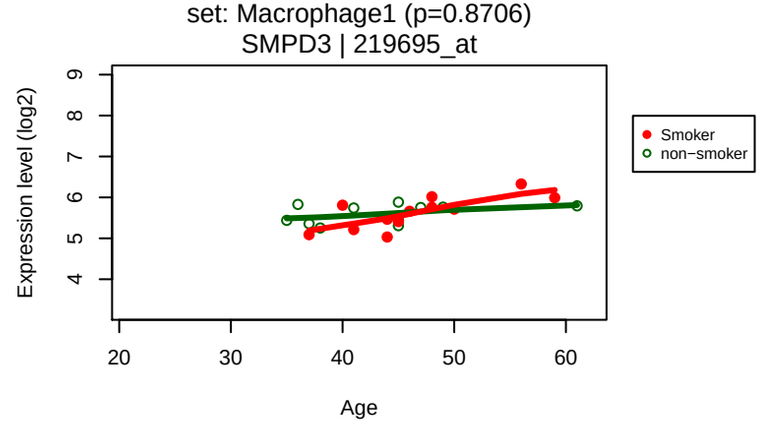

set: SmallAirway1 (p=0.6808)
SMPD3 | 219695_at
Expression level (log2)
Age
Smoker
non−smoker

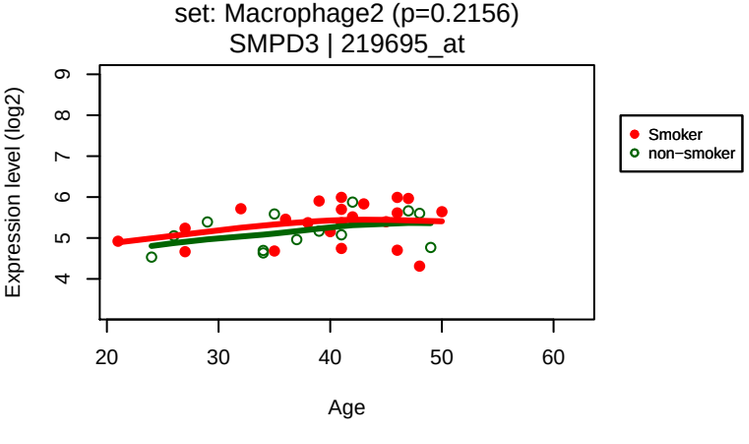

set: SmallAirway2 (p=0.1322)
SMPD3 | 219695_at
Expression level (log2)
Age
Smoker
non−smoker

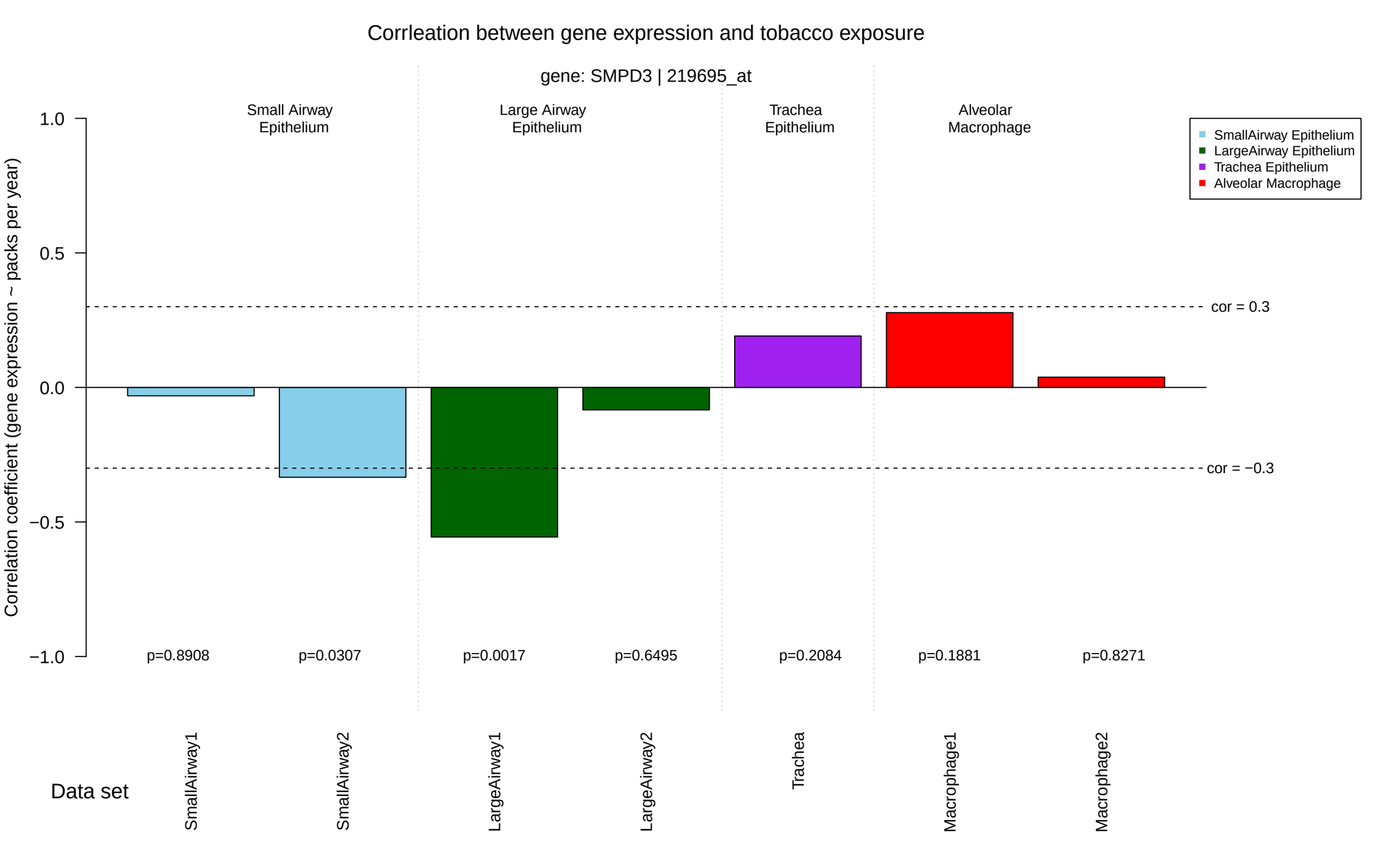

set: LargeAirway1 (p=0.0868)
SMPD3 | 219695_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9108)
SMPD3 | 219695_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1363)
SMPD3 | 219695_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8706)
SMPD3 | 219695_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2156)
SMPD3 | 219695_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMPD3 | 219695_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8908
p=0.0307
p=0.0017
p=0.6495
p=0.2084
p=0.1881
p=0.8271
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2